

Member 2 Report – Body Segmentation & Pose Detection

Objective

The objective of this task was to implement body segmentation and pose detection techniques that can be used as preprocessing stages in virtual try-on systems. The experiments focused on separating the human body from the background and detecting body landmarks required for body analysis and garment alignment.

Introduction

Body segmentation and pose detection are fundamental components of computer vision applications such as virtual try-on systems, human activity recognition, motion tracking, and augmented reality.

Body segmentation isolates the person from the background, while pose detection identifies important body landmarks such as shoulders, elbows, hips, knees, and ankles. These landmarks provide information about body structure and posture, which can be used for garment fitting and alignment.

In this experiment, MediaPipe Selfie Segmentation and MediaPipe Pose were used to perform body segmentation and pose estimation.

Tools and Technologies Used

MediaPipe

MediaPipe is an open-source framework developed by Google that provides real-time machine learning solutions for computer vision applications.

MediaPipe Selfie Segmentation

Used for separating the human body from the background.

MediaPipe Pose

Used for detecting body landmarks and generating skeletal structures.

Detectron2 (Research Study)

Detectron2 was studied as an advanced segmentation framework and compared with MediaPipe in terms of accuracy and computational requirements.

OpenCV

Used for image processing and visualization.

Python

Used for implementing segmentation and pose detection algorithms.

Experiment 1: Body Segmentation

Objective

To isolate the human body from the background and evaluate segmentation quality.

Methodology

1. Input image was loaded into the system.
2. MediaPipe Selfie Segmentation model was applied.
3. The segmentation mask was generated.
4. Background pixels were removed.
5. Segmented output was displayed and analyzed.

Original Image



Results

The model successfully separated the person from the background while preserving important body regions.

Observations

- High-quality segmentation was obtained for front-facing images.
- Body boundaries were accurately detected.
- Minor inaccuracies were observed near hair and shoulder regions.
- Performance was suitable for virtual try-on preprocessing tasks.

Body Segmentation Output



Experiment 2: Pose Detection

Objective

To identify body landmarks and generate a skeletal representation of the human body.

Methodology

1. Input image was processed using MediaPipe Pose.
2. Body landmarks were extracted.
3. Skeletal connections were drawn between landmarks.
4. Landmark positions were visualized and analyzed

Detected Landmarks

The model detected 33 body landmarks, including:

- Nose
- Eyes
- Ears
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

Results

The pose detection model successfully generated a complete skeletal structure and accurately identified major body joints.



Observations

- Shoulder and hip landmarks were detected accurately.
- Skeleton generation was stable and consistent.
- Body posture was correctly represented.
- Suitable for body measurement and garment alignment applications.

Accuracy Evaluation

The models were tested on multiple body poses.

Body Pose	Segmentation Accuracy	Pose Detection Accuracy
Front Pose	High	High
Side Pose	Medium	High
Arms Raised	High	High
Sitting Pose	Medium	Medium
Walking Pose	Medium	High

Analysis

Front-facing images produced the highest accuracy for both segmentation and pose detection.

Complex poses slightly affected segmentation quality due to body part overlap. However, pose detection continued to identify major landmarks effectively.

MediaPipe vs Detectron2 Comparison

Feature	MediaPipe	Detectron2
Processing Speed	Fast	Moderate
Real-Time Capability	Yes	Limited
Segmentation Accuracy	Good	Very High
Hardware Requirement	Low	High
Ease of Implementation	Easy	Moderate
Virtual Try-On Suitability	High	High

Comparison Summary

MediaPipe provides lightweight and real-time performance suitable for virtual try-on applications. Detectron2 offers superior segmentation accuracy but requires greater computational resources.

Experimental Outputs

The following outputs were generated during experimentation:

- 1. Original Image
- 2. Segmentation Output
- 3. Pose Detection Output
- 4. Skeleton Visualization
- 5. Landmark Detection Results

Screenshots of these outputs were collected and documented.

Conclusion

Body segmentation and pose detection were successfully implemented using MediaPipe. The segmentation model effectively separated the human body from the background, while the pose detection model accurately identified body landmarks and skeletal structures. The results demonstrate that these techniques can serve as reliable preprocessing stages for virtual try-on systems and other computer vision applications.